#4 Estimate uncertainty (*P*), process noise (*Q*), and measurement noise (*R*)
#5 Prediction step.
#6 Update step.

Figure 3.10 shows the output of the code in lisiting 3.5, plotting the true temperature, the mean of the five temperature sensors, and the temperature estimated by the Kalman filter at each time step. At each step, the filtered estimate is closer to the actual temperature than the mean of the sensor readings.

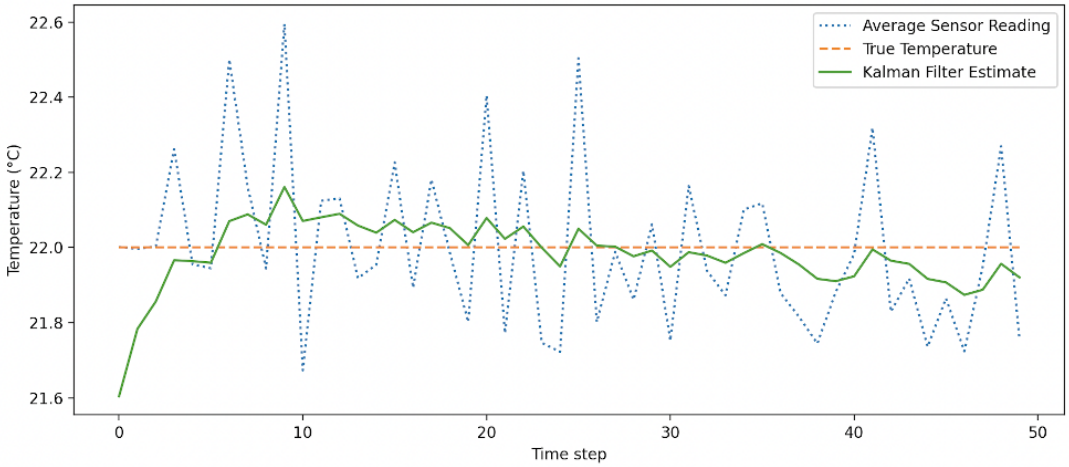


Figure 3.10 Output of the simple Kalman filter running against five noisy temperature sensors as shown in listing 3.5.

3.6 Scalability and device management

Each sensor that you deploy adds multiple management dimensions that compound exponentially as your deployments grow. Deploying the first ten sensors around my home was easy, but as I add more sensors even in the small scale of my home digital twin, managing them manually becomes impossible. I need to keep track of which sensor is deployed where, what version of firmware each is running, and when batteries were last changed. The challenge of the type of large scale IoT deployments required by an industrial scale digital twin is in keeping sensors working reliably at scale. Following are the challenges that you will face as you scale up sensor deployments for your digital twin.